



Derivative Using the Limit Definition

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Apply the limit definition of the derivative to find $f'(x)$
- Simplify difference quotients by expanding and combining like terms
- Evaluate limits as h approaches zero to obtain the derivative

Use the limit definition of the derivative, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, to find the derivative of each function.

1. Find the derivative using the limit definition.

$$f(x) = 5x + 3$$

Answer: _____

2. Find the derivative using the limit definition.

$$f(x) = 3x^2 - 2x$$

Answer: _____

3. Find the derivative using the limit definition.

$$f(x) = 7x - 4$$

Answer: _____

4. Find the derivative using the limit definition.

$$f(x) = x^2$$

Answer: _____

5. Find the derivative using the limit definition.

$$f(x) = x^2 + 4x$$

Answer: _____

6. Find the derivative using the limit definition.

$$f(x) = 2x^2 - 5x + 1$$

Answer: _____

7. Find the derivative using the limit definition.

$$f(x) = x^3$$

Answer: _____



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8. Find the derivative using the limit definition.

$$f(x) = \frac{1}{x}$$

Answer: _____

9. Find the derivative using the limit definition.

$$f(x) = \sqrt{x}$$

Answer: _____

10. Find the derivative using the limit definition.

$$f(x) = 4x^2 + 3$$

Answer: _____





Encourage students to carefully expand $f(x+h)$ and distribute negative signs before simplifying the difference quotient.

Solutions

1. Find the derivative using the limit definition.

$$f(x) = 5x + 3$$

- Compute $f(x+h)$ by replacing x with $x+h$, giving $5(x+h) + 3$.
- Form the difference quotient: $[5(x+h) + 3 - (5x + 3)] / h$.
- Distribute and simplify the numerator to get $5h/h$.
- Cancel h and take the limit as h approaches 0 to get 5.

Answer: $f'(x) = 5$

2. Find the derivative using the limit definition.

$$f(x) = 3x^2 - 2x$$

- Compute $f(x+h) = 3(x+h)^2 - 2(x+h) = 3x^2 + 6xh + 3h^2 - 2x - 2h$.
- Subtract $f(x)$ to get $6xh + 3h^2 - 2h$ in the numerator.
- Factor h from the numerator and divide by h to obtain $6x + 3h - 2$.
- Take the limit as h approaches 0 to get $6x - 2$.

Answer: $f'(x) = 6x - 2$

3. Find the derivative using the limit definition.

$$f(x) = 7x - 4$$

- Compute $f(x+h) = 7(x+h) - 4 = 7x + 7h - 4$.
- Form the difference quotient: $(7x + 7h - 4 - 7x + 4) / h = 7h/h$.
- Simplify the expression to 7.
- Take the limit as h approaches 0 to get 7.

Answer: $f'(x) = 7$

4. Find the derivative using the limit definition.

$$f(x) = x^2$$

- Compute $f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$.
- Subtract $f(x) = x^2$ to get $2xh + h^2$ in the numerator.
- Divide by h to obtain $2x + h$.
- Take the limit as h approaches 0 to get $2x$.

Answer: $f'(x) = 2x$



5. Find the derivative using the limit definition.

$$f(x) = x^2 + 4x$$

- Compute $f(x+h) = (x+h)^2 + 4(x+h) = x^2 + 2xh + h^2 + 4x + 4h$.
- Subtract $f(x) = x^2 + 4x$ to get $2xh + h^2 + 4h$.
- Factor h and divide to obtain $2x + h + 4$.
- Take the limit as h approaches 0 to get $2x + 4$.

Answer: $f'(x) = 2x + 4$

6. Find the derivative using the limit definition.

$$f(x) = 2x^2 - 5x + 1$$

- Compute $f(x+h) = 2(x+h)^2 - 5(x+h) + 1 = 2x^2 + 4xh + 2h^2 - 5x - 5h + 1$.
- Subtract $f(x) = 2x^2 - 5x + 1$ to leave $4xh + 2h^2 - 5h$ in the numerator.
- Factor h and divide by h to obtain $4x + 2h - 5$.
- Take the limit as h approaches 0 to get $4x - 5$.

Answer: $f'(x) = 4x - 5$

7. Find the derivative using the limit definition.

$$f(x) = x^3$$

- Compute $f(x+h) = (x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$.
- Subtract $f(x) = x^3$ to get $3x^2h + 3xh^2 + h^3$.
- Factor h and divide by h to obtain $3x^2 + 3xh + h^2$.
- Take the limit as h approaches 0 to get $3x^2$.

Answer: $f'(x) = 3x^2$

8. Find the derivative using the limit definition.

$$f(x) = \frac{1}{x}$$

- Compute $f(x+h) = 1/(x+h)$.
- Form the difference quotient: $[1/(x+h) - 1/x] / h = [-h / (x(x+h))] / h$.
- Simplify by canceling h to get $-1 / (x(x+h))$.
- Take the limit as h approaches 0 to obtain $-1/x^2$.

Answer: $f'(x) = -\frac{1}{x^2}$

9. Find the derivative using the limit definition.

$$f(x) = \sqrt{x}$$

- Form the difference quotient: $[\sqrt{x+h} - \sqrt{x}] / h$.
- Multiply numerator and denominator by the conjugate $\sqrt{x+h} + \sqrt{x}$.
- Simplify the numerator to h , giving $h / [h(\sqrt{x+h} + \sqrt{x})]$.
- Cancel h and take the limit as h approaches 0 to get $1/(2\sqrt{x})$.

Answer: $f'(x) = \frac{1}{2\sqrt{x}}$



10. Find the derivative using the limit definition.

$$f(x) = 4x^2 + 3$$

→ Compute $f(x+h) = 4(x+h)^2 + 3 = 4x^2 + 8xh + 4h^2 + 3$.

→ Subtract $f(x) = 4x^2 + 3$ to get $8xh + 4h^2$.

→ Factor h and divide by h to obtain $8x + 4h$.

→ Take the limit as h approaches 0 to get $8x$.

Answer: $f'(x) = 8x$

